SHOWTIME
BUSINESS REPORT

Predictive Modelling Coded Project

By

Rajesh Mambulli

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1. Context

An over-the-top (OTT) media service is a media service offered directly to viewers via the internet. The term is most synonymous with subscription-based video-on-demand services that offer access to film and television content, including existing series acquired from other producers, as well as original content produced specifically for the service. They are typically accessed via websites on personal computers, apps on smartphones and tablets, or televisions with integrated Smart TV platforms.

Presently, OTT services are at a relatively nascent stage and are widely accepted as a trending technology across the globe. With the increasing change in customers' social behaviour, which is shifting from traditional subscriptions to broadcasting services and OTT on-demand video and music subscriptions every year, OTT streaming is expected to grow at a very fast pace. The global OTT market size was valued at $121.61 billion in 2019 and is projected to reach $1,039.03 billion by 2027, growing at a CAGR of 29.4% from 2020 to 2027. The shift from television to OTT services for entertainment is driven by benefits such as on-demand services, ease of access, and access to better networks and digital connectivity.

With the outbreak of COVID19, OTT services are striving to meet the growing entertainment appetite of viewers, with some platforms already experiencing a 46% increase in consumption and subscriber count as viewers seek fresh content. With innovations and advanced transformations, which will enable the customers to access everything they want in a single space, OTT platforms across the world are expected to increasingly attract subscribers on a concurrent basis.

2. Objective

ShowTime is an OTT service provider and offers a wide variety of content (movies, web shows, etc.) for its users. They want to determine the driver variables for first-day content viewership so that they can take necessary measures to improve the viewership of the content on their platform. Some of the reasons for the decline in viewership of content would be the decline in the number of people coming to the platform, decreased marketing spends, content timing clashes, weekends, and holidays, etc. They have hired you as a Data Scientist, shared the data of the current content in their platform, and asked you to analyse the data and come up with a linear regression model to determine the driving factors for first-day viewership.

3. Data Description

The data contains the different factors to analyse for the content. The detailed data dictionary is given below.

4. Data Dictionary:

- visitors: Average number of visitors, in millions, to the platform in the past week
- ad_impressions: Number of ad impressions, in millions, across all ad campaigns for the content (running and completed)
- major_sports_event: Any major sports event on the day
- genre: Genre of the content
- dayofweek: Day of the release of the content
- season: Season of the release of the content
- views_trailer: Number of views, in millions, of the content trailer
- views_content: Number of first-day views, in millions, of the content

5. Data Overview through Python

Data which is stored in csv file is imported to Jupyter Notebook with the help of pandas, NumPy libraries and used matplotlib and seaborn libraries to help with data visualization. Dataset is stored in python variable named data.

Understanding the shape of the dataset.

This dataset contains 1000 rows and 8 columns.

	visitors	ad_impressions	major_sports_event	genre	dayofweek	season	views_trailer	views_content
0	1.67	1113.81	0	Horror	Wednesday	Spring	56.70	0.51
1	1.46	1498.41	1	Thriller	Friday	Fall	52.69	0.32
2	1.47	1079.19	1	Thriller	Wednesday	Fall	48.74	0.39
3	1.85	1342.77	1	Sci-Fi	Friday	Fall	49.81	0.44
4	1.46	1498.41	0	Sci-Fi	Sunday	Winter	55.83	0.46

Figure 1: Top 5 rows of data

Checking the data types of data.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 8 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   visitors            1000 non-null   float64
 1   ad_impressions      1000 non-null   float64
 2   major_sports_event  1000 non-null   int64
 3   genre               1000 non-null   object
 4   dayofweek           1000 non-null   object
 5   season              1000 non-null   object
 6   views_trailer       1000 non-null   float64
 7   views_content       1000 non-null   float64
dtypes: float64(4), int64(1), object(3)
memory usage: 62.6+ KB
```

Figure 2: Data Types of attributes

All the Numerical attributes are in either int or float datatype which is desired. And remaining categories are string object type.

This concludes there is no need to change the data types of any attributes listed in the dataset.

5.1 Checking for missing values in the data.

```
visitors              0
ad_impressions        0
major_sports_event    0
genre                 0
dayofweek             0
season                0
views_trailer         0
views_content         0
dtype: int64
```

- No Null values in the data

Figure 3: Null checks in the data

5.2 Checking for duplicate values in the data.

```
0
```

- No duplicates in the dataset.

Figure 4: Duplicate checks in the data

5.3 Checking the statistical summary of the numerical data.

	count	mean	std	min	25%	50%	75%	max
visitors	1000.0	1.70429	0.231973	1.25	1.5500	1.70	1.830	2.34
ad_impressions	1000.0	1434.71229	289.534834	1010.87	1210.3300	1383.58	1623.670	2424.20
major_sports_event	1000.0	0.40000	0.490143	0.00	0.0000	0.00	1.000	1.00
views_trailer	1000.0	66.91559	35.001080	30.08	50.9475	53.96	57.755	199.92
views_content	1000.0	0.47340	0.105914	0.22	0.4000	0.45	0.520	0.89

Figure 5: Statistical summary of data

Visitors

The dataset consists of 1000 entries with the number of visitors. On average, there are approximately 1.704 visitors, with a standard deviation of 0.232. The minimum number of visitors recorded is 1.25, and the maximum is 2.34. The data is fairly cantered around 1.7, as evidenced by the median value.

Ad Impressions

This dataset shows the number of ad impressions with a mean value of 1434.71 and a standard deviation of 289.53. This suggests that while most impressions are around the average, there is a considerable spread. The minimum number of impressions is 1010.87, while the highest reaches 2424.20. Quartiles indicate that 25% of the data falls below 1210.33, and 75% falls below 1623.67.

Major Sports Event

This categorical variable indicates whether a major sports event is taking place. The mean value of 0.40, combined with a standard deviation of 0.49, suggests that sports events occur less frequently than not. The minimum and median values are 0, while the maximum is 1, indicating some instances of such events.

Views Trailer

The number of trailer views averages 66.92, with a standard deviation of 35.00, which shows high variability. The dataset ranges from 30.08 to 199.92 views. The lower quartile is around 50.95, and the upper quartile is approximately 57.76, indicating that most of the data is concentrated between these values.

Views Content

For views of the main content, the average value is 0.47 with a standard deviation of 0.11, The minimum value is 0.22 and the maximum is 0.89, with the median at 0.45. Quartiles indicate that 25% of the data is below 0.40 and 75% is below 0.52.

6.0 Univariate Analysis

Let us check the distribution for numerical columns.

6.1 Observations on visitors

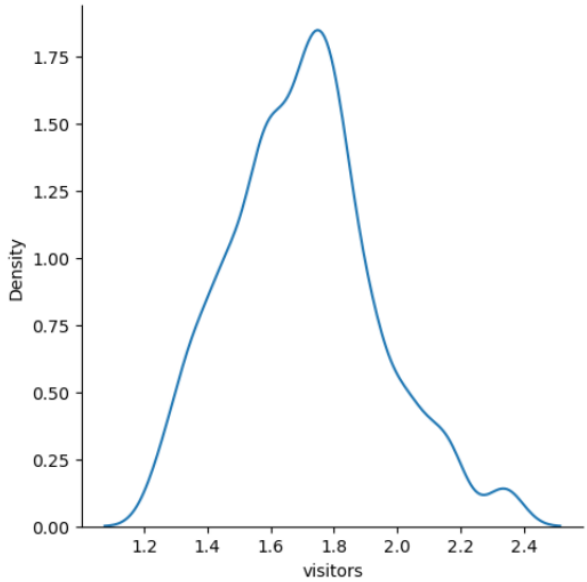

Figure 6: Distribution of visitor's attribute

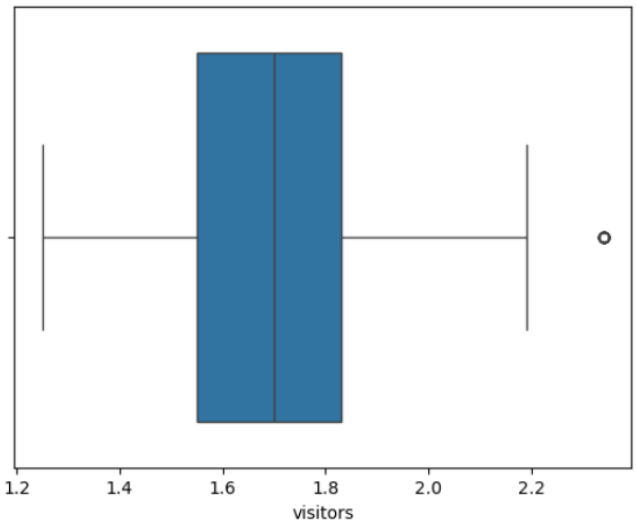

Figure 7: outliers in the visitors attribute

From the above 2 plots below mentioned are the observations

- The distribution is normal
- There is one outlier present in this visitor's column.
- Values above 2.2 million are being represented as outliers in the boxplot.

6.2 Observations on ad_impressions

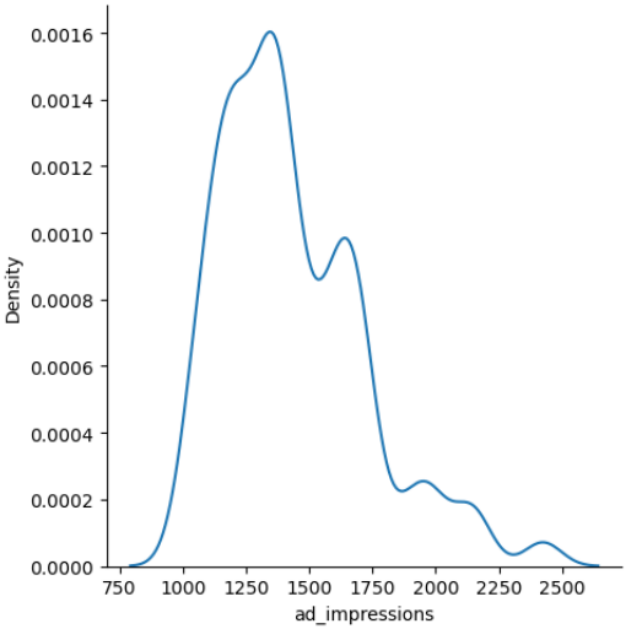

Figure 8: Distribution of ad_impressions attribute

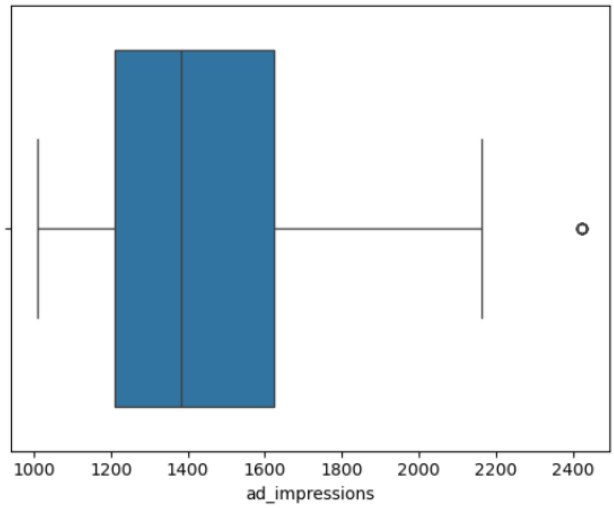

Figure 9: Outliers in ad_impressions attribute

From the above 2 plots below mentioned are the observations

- The distribution is skewed towards the right.
- There is one outlier present in this ad_impressions column.

6.3 Observations on major_sports_event

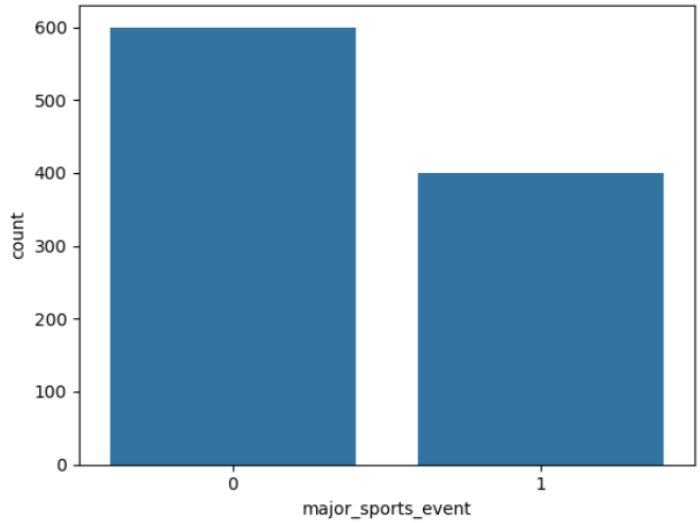

Figure 10: Distribution of Major_sports_event attribute

From the above plot below mentioned are the observations

Data contains 40% Major Sports event data and 60% No Major Sports event data on the day the content released.

6.4 Observations on views_trailer

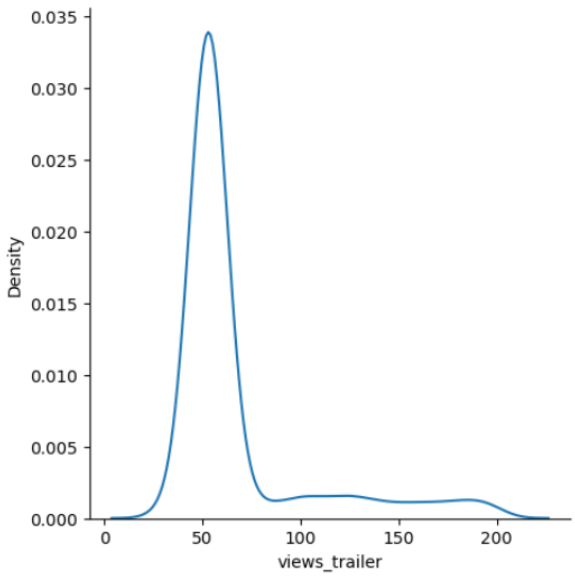

Figure 11: Distribution of views_trailer attribute

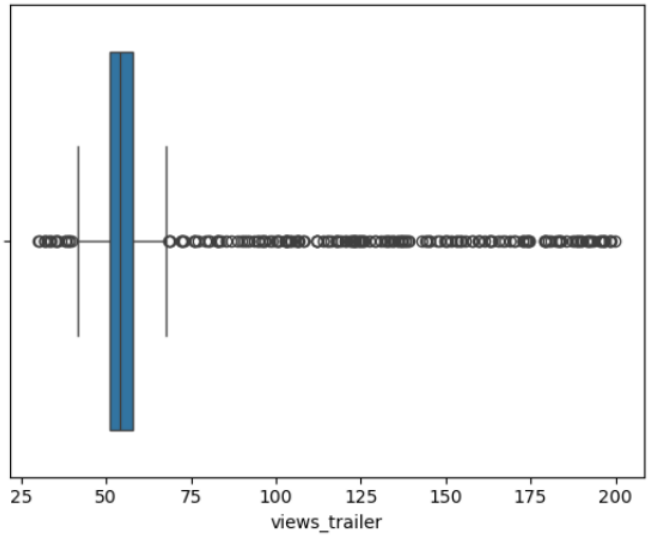

Figure 12: Outliers in views_trailer attribute

From the above plot below mentioned are the observations

- data is skewed towards right.
- There are many outliers in this attribute.

6.5 Observations on views_content

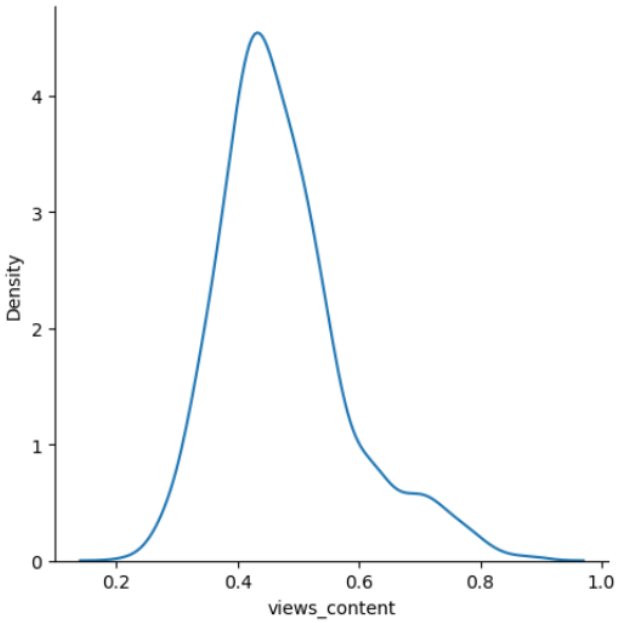

Figure 13: Distribution of views_content attribute

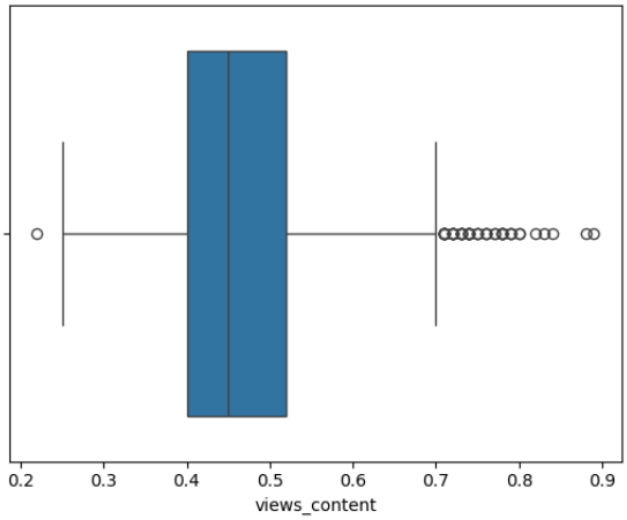

Figure 14: outliers of view_content attribute

From the above plot below mentioned are the observations

- Data is slightly skewed towards right.
- There are some outliers in this attribute.
- We will see will we Require Outlier treatment using IQR method.

6.6 Observations on genre

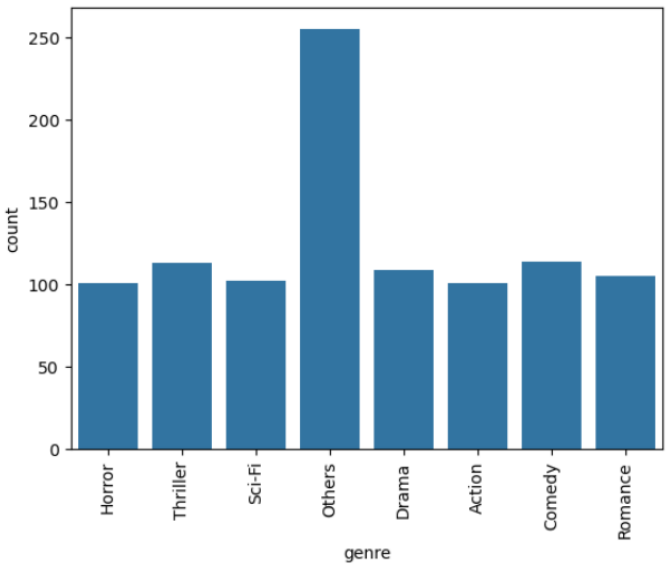

Figure 15: Distribution of genre

- All the genres are having more or less equal count of content. But majority is categorized as others
- Thriller and Comedy Genres have most content next to other genres.

6.7 Observations on dayofweek

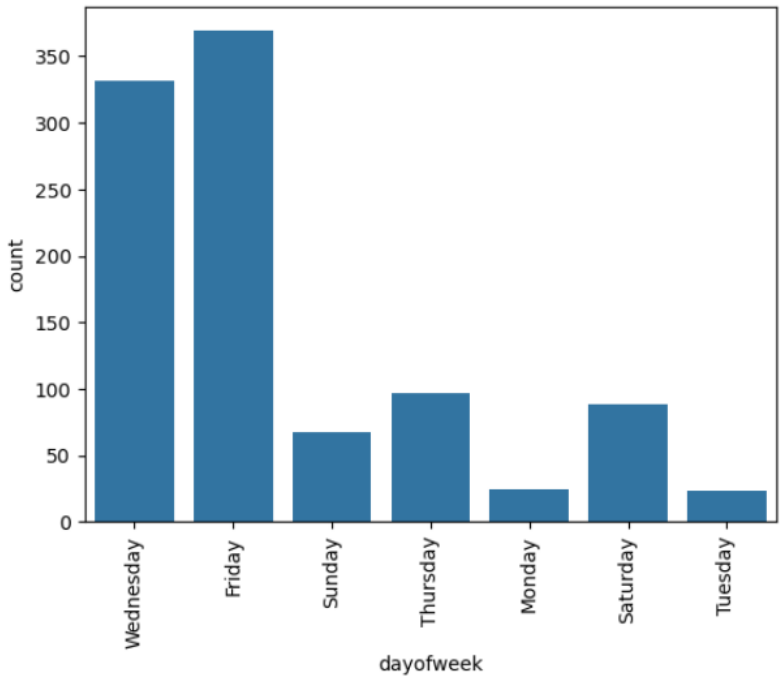

Figure 16: Distribution of day of week attribute

- Most of the content released are on Friday and Wednesday.
- Monday and Tuesday have a smaller number of contents released.

6.7 Observations on season

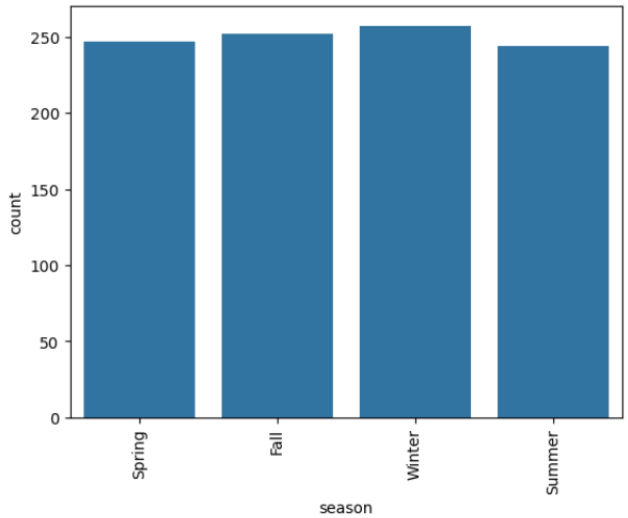

Figure 17: Distribution of season attribute

- Comparatively speaking all the Seasons has equal amounts of content releases.
- Winter has most number of releases.

7.0 Bivariate Analysis

Figure 18: Heat map of numerical data.

Below mentioned are the observations.

- views_content shows high correlation with views_trailer. This indicates that people who sees trailer sees content on the same day.
- major_sports_event shows negative correlation with views_content. This indicates that on sports day content view on OTT is getting negatively impacted.
- views_content has positive correlation with visitors.
- ad_impressions has positive correlation but not that compared to other attributes in the data.

7.1 Observations on views_content and views_trailer.

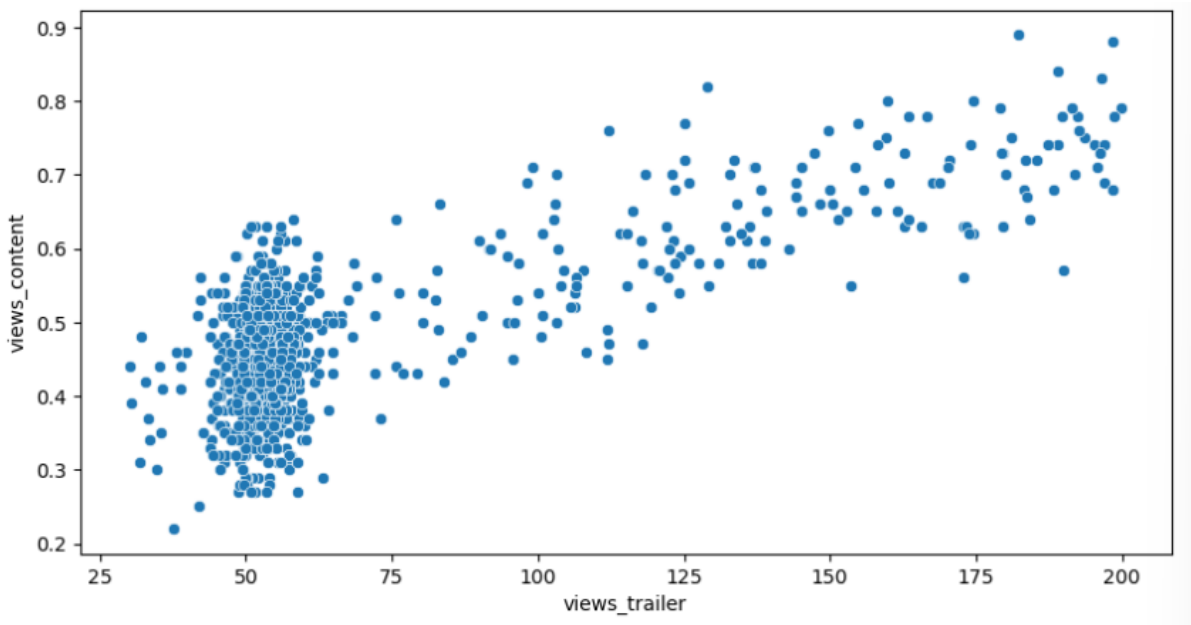

Figure 19: relation between views_content and views_trailer

- A positive correlation or an increasing trend can be clearly observed between views_trailer and views_content.

7.2 Observations on views_content and dayofrelease

dayofweek	views_content
Friday	164.83
Monday	11.23
Saturday	43.82
Sunday	32.44
Thursday	45.65
Tuesday	11.22
Wednesday	164.21

Figure 20: Summary of day of release and views_content

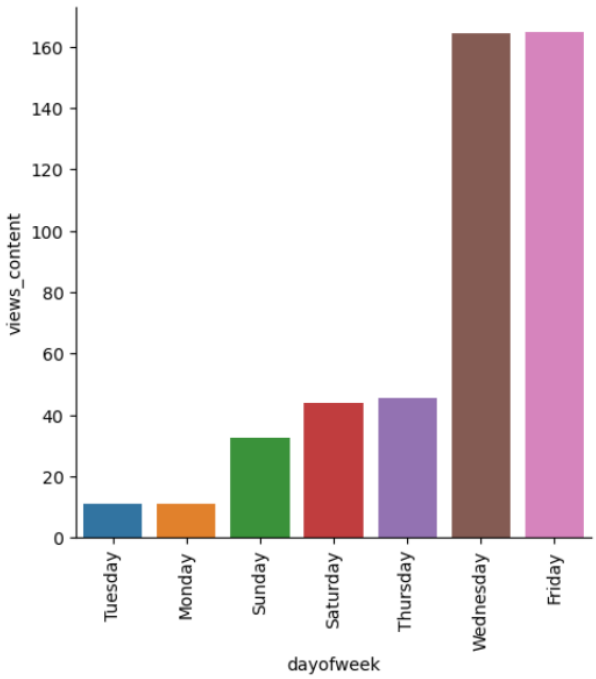

Figure 21: Day of week vs views_content

Observations on above 2 graphs

- First day views_count is more for the content which release on Wednesday and Friday.
- Content released on Friday has more total number of First day views.
- Content released on Wednesday is in next in line beside Friday wrt to First day total views.
- Content released on Tuesday has least total number of First day views compared to all other days.

7.3 Observations on views_content and season

season	views_content
Fall	112.23
Spring	115.39
Summer	121.22
Winter	124.56

Figure 22: summary of views_content and season

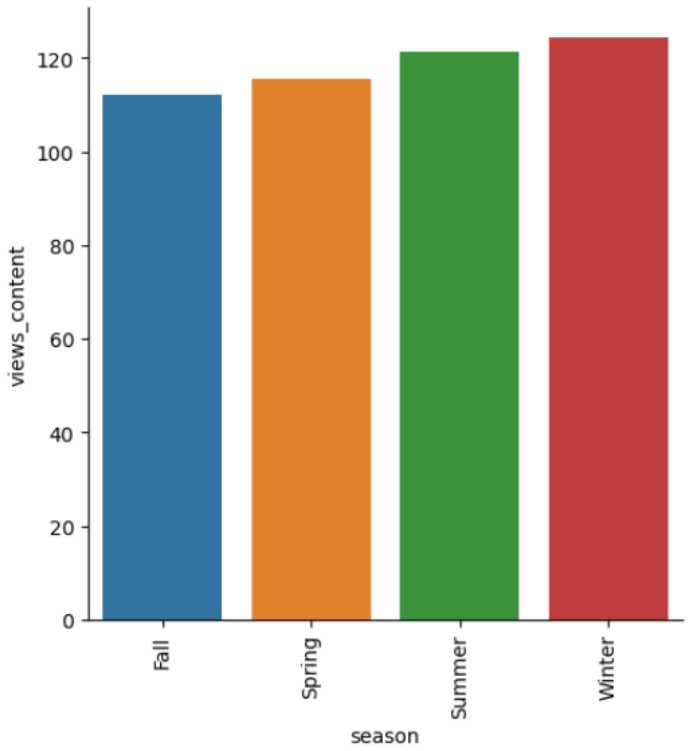

Figure 23: Season vs views_content

Observations from above 2 graphs

- Content released in Winter season has higher First day viewership
- Content released in summer season is higher First day viewership next to winter.
- Content released in Fall season has least total First day viewership compared to all the other seasons.

8.0 Finding Outliers

Finding outliers on numerical columns.

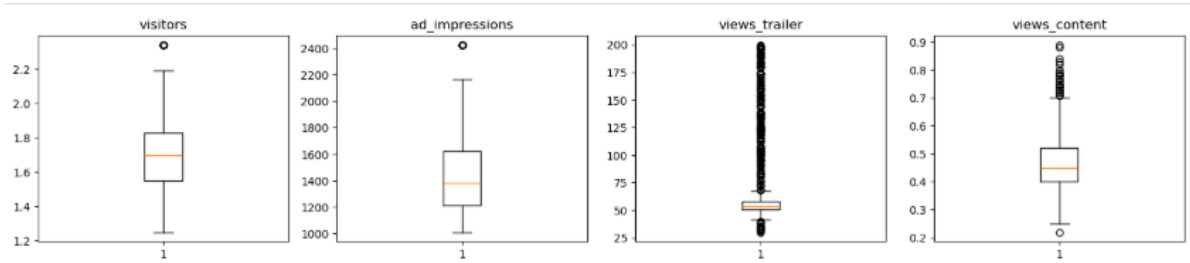

Figure 24: Outliers in Numerical data.

Each of the 4 numerical attributes contain outliers but these are real values which are true in nature in Realtime. So, treating them would be manipulating the data which in turn effects the predictive model later half of the project.

Percentages of Outliers in each column

```
visitors           2.0
ad_impressions     1.3
views_trailer     18.9
views_content      4.7
dtype: float64
```

Figure 25: Percentage of outliers in numerical columns

- There are quite a lot outliers in the data.
- However, we will not treat them as they are proper values.

9.0 Data Preparation for Modelling

```
   visitors  ad_impressions  major_sports_event     genre  dayofweek  season  \
0      1.67         1113.81                   0    Horror  Wednesday  Spring
1      1.46         1498.41                   1  Thriller     Friday    Fall
2      1.47         1079.19                   1  Thriller  Wednesday    Fall
3      1.85         1342.77                   1    Sci-Fi     Friday    Fall
4      1.46         1498.41                   0    Sci-Fi     Sunday  Winter

   views_trailer
0          56.70
1          52.69
2          48.74
3          49.81
4          55.83
0    0.51
1    0.32
2    0.39
3    0.44
4    0.46
Name: views_content, dtype: float64
```

Figure 26: Data split evidence

Divided the X and Y variables in the data X being all the other attributes except the views_content field.

Adding constant to the X data and creating dummy variables for categorical variables.

	const	visitors	ad_impressions	major_sports_event	views_trailer	genre_Comedy	genre_Drama	genre_Horror	genre
0	1.0	1.67	1113.81	0	56.70	False	False	True	
1	1.0	1.46	1498.41	1	52.69	False	False	False	
2	1.0	1.47	1079.19	1	48.74	False	False	False	
3	1.0	1.85	1342.77	1	49.81	False	False	False	
4	1.0	1.46	1498.41	0	55.83	False	False	False	

5 rows × 21 columns

Figure 27: Evidence of adding constant and adding dummy variables

	const	visitors	ad_impressions	major_sports_event	views_trailer	genre_Comedy	genre_Drama	genre_Horror	genre
0	1.0	1.67	1113.81	0.0	56.70	0.0	0.0	1.0	
1	1.0	1.46	1498.41	1.0	52.69	0.0	0.0	0.0	
2	1.0	1.47	1079.19	1.0	48.74	0.0	0.0	0.0	
3	1.0	1.85	1342.77	1.0	49.81	0.0	0.0	0.0	
4	1.0	1.46	1498.41	0.0	55.83	0.0	0.0	0.0	

5 rows × 21 columns

Figure 28: Converting Boolean values to Float values.

Replacing all the Boolean variables which were created above to the Float datatype.

Splitting the data into Train and Test data based on rule 70 and 30 respectively.

```
Number of rows in train data = 700
Number of rows in test data = 300
```

Figure 29: Evidence of Test and Train split of Data

10.0 Model Building - Linear Regression

```
                            OLS Regression Results
==============================================================================
Dep. Variable:          views_content   R-squared:                       0.792
Model:                            OLS   Adj. R-squared:                  0.785
Method:                 Least Squares   F-statistic:                     129.0
Date:                Sun, 15 Dec 2024   Prob (F-statistic):          1.32e-215
Time:                        19:54:07   Log-Likelihood:                 1124.6
No. Observations:                 700   AIC:                            -2207.
Df Residuals:                     679   BIC:                            -2112.
Df Model:                          20
Covariance Type:            nonrobust
==========================================================================================
                           coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                    0.0602      0.019      3.235      0.001       0.024       0.097
visitors                 0.1295      0.008     16.398      0.000       0.114       0.145
ad_impressions        3.623e-06   6.58e-06      0.551      0.582    -9.3e-06    1.65e-05
major_sports_event      -0.0603      0.004    -15.284      0.000      -0.068      -0.053
views_trailer            0.0023   5.52e-05     42.193      0.000       0.002       0.002
genre_Comedy             0.0094      0.008      1.172      0.241      -0.006       0.025
genre_Drama              0.0126      0.008      1.554      0.121      -0.003       0.029
genre_Horror             0.0099      0.008      1.207      0.228      -0.006       0.026
genre_Others             0.0063      0.007      0.897      0.370      -0.008       0.020
genre_Romance            0.0006      0.008      0.065      0.948      -0.016       0.017
genre_Sci-Fi             0.0131      0.008      1.599      0.110      -0.003       0.029
genre_Thriller           0.0087      0.008      1.079      0.281      -0.007       0.025
dayofweek_Monday         0.0337      0.012      2.848      0.005       0.010       0.057
dayofweek_Saturday       0.0579      0.007      8.094      0.000       0.044       0.072
dayofweek_Sunday         0.0363      0.008      4.639      0.000       0.021       0.052
dayofweek_Thursday       0.0173      0.007      2.558      0.011       0.004       0.031
dayofweek_Tuesday        0.0228      0.014      1.665      0.096      -0.004       0.050
dayofweek_Wednesday      0.0474      0.004     10.549      0.000       0.039       0.056
season_Spring            0.0226      0.005      4.224      0.000       0.012       0.033
season_Summer            0.0442      0.005      8.111      0.000       0.034       0.055
season_Winter            0.0272      0.005      5.096      0.000       0.017       0.038
==============================================================================
Omnibus:                        3.850   Durbin-Watson:                   2.004
Prob(Omnibus):                  0.146   Jarque-Bera (JB):                3.722
Skew:                           0.143   Prob(JB):                        0.156
Kurtosis:                       3.215   Cond. No.                     1.67e+04
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.67e+04. This might indicate that there are
strong multicollinearity or other numerical problems.
```

Figure 30: Regression Summary

R-squared:

* The value for R-squared is 0.792, which is good.

Adjusted. R-squared:

* The value for adj. R-squared is 0.785, which is good.

Const coefficient:

* The value for const coefficient is 0.0602

10.1 Model Performance Check

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.04853	0.038197	0.791616	0.785162	8.55644

Figure 31: Model Performance Summary of Train data

Observations

- The training R2 is 0.79, so the model is not underfitting
- The train and test RMSE and MAE are comparable, so the model is not overfitting either
- MAE suggests that the model can predict First day viewership within a mean error of 0.034 on the test data
- MAPE of 12.6 on the test data means that we are able to predict within 8.55% of the First day viewership.

10.1 Checking Linear Regression Assumptions

We will be checking the following Linear Regression assumptions:

1. **No Multicollinearity**
2. **Linearity of variables**
3. **Independence of error terms**
4. **Normality of error terms**
5. **No Heteroscedasticity**

10.1.1 TEST FOR MULTICOLLINEARITY

General Rule of thumb:

If VIF is between 1 and 5, then there is low multicollinearity.

If VIF is between 5 and 10, we say there is moderate multicollinearity.

If VIF is exceeding 10, it shows signs of high multicollinearity.

	feature	VIF
0	const	99.679317
1	visitors	1.027837
2	ad_impressions	1.029390
3	major_sports_event	1.065689
4	views_trailer	1.023551
5	genre_Comedy	1.917635
6	genre_Drama	1.926699
7	genre_Horror	1.904460
8	genre_Others	2.573779
9	genre_Romance	1.753525
10	genre_Sci-Fi	1.863473
11	genre_Thriller	1.921001
12	dayofweek_Monday	1.063551
13	dayofweek_Saturday	1.155744
14	dayofweek_Sunday	1.150409
15	dayofweek_Thursday	1.169870
16	dayofweek_Tuesday	1.062793
17	dayofweek_Wednesday	1.315231
18	season_Spring	1.541591
19	season_Summer	1.568240
20	season_Winter	1.570338

Figure 32: VIF of Features

Observations on above image.

- There are no multiple columns with very high VIF values, indicating no presence of multicollinearity
- We will systematically drop numerical columns with VIF > 5
- We will ignore the VIF values for dummy variables and the constant (intercept)

10.1.2 Dealing with high p-value variables

After dropping of all the variables whose p value greater than 0.05

```
                            OLS Regression Results
==============================================================================
Dep. Variable:          views_content   R-squared:                       0.789
Model:                            OLS   Adj. R-squared:                  0.786
Method:                 Least Squares   F-statistic:                     233.8
Date:                Sun, 15 Dec 2024   Prob (F-statistic):          7.03e-224
Time:                        19:54:07   Log-Likelihood:                 1120.2
No. Observations:                 700   AIC:                            -2216.
Df Residuals:                     688   BIC:                            -2162.
Df Model:                          11
Covariance Type:            nonrobust
=========================================================================================
                            coef    std err          t      P>|t|      [0.025      0.975]
-----------------------------------------------------------------------------------------
const                     0.0747      0.015      5.110      0.000       0.046       0.103
visitors                  0.1291      0.008     16.440      0.000       0.114       0.145
major_sports_event       -0.0606      0.004    -15.611      0.000      -0.068      -0.053
views_trailer             0.0023    5.5e-05     42.414      0.000       0.002       0.002
dayofweek_Monday          0.0321      0.012      2.731      0.006       0.009       0.055
dayofweek_Saturday        0.0570      0.007      8.042      0.000       0.043       0.071
dayofweek_Sunday          0.0344      0.008      4.456      0.000       0.019       0.050
dayofweek_Thursday        0.0154      0.007      2.307      0.021       0.002       0.029
dayofweek_Wednesday       0.0465      0.004     10.532      0.000       0.038       0.055
season_Spring             0.0226      0.005      4.259      0.000       0.012       0.033
season_Summer             0.0434      0.005      8.112      0.000       0.033       0.054
season_Winter             0.0282      0.005      5.362      0.000       0.018       0.039
==============================================================================
Omnibus:                        3.254   Durbin-Watson:                   1.996
Prob(Omnibus):                  0.196   Jarque-Bera (JB):                3.077
Skew:                           0.139   Prob(JB):                        0.215
Kurtosis:                       3.168   Cond. No.                         662.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

Figure 33: Regression Summary

Training Performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.048841	0.038385	0.788937	0.785251	8.595246

Figure 34: Train Data Performance

Test Performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.051109	0.041299	0.761753	0.751792	9.177097

Figure 35: Test Data Performance

Observations

- Now no feature has p-value greater than 0.05, so we'll consider the features in *x_train2* as the final set of predictor variables and *olsmod2* as the final model to move forward with
- Now adjusted R-squared is 0.785, i.e., our model is able to explain ~79% of the variance
- The adjusted R-squared in *olsmod1* (where we considered the variables without multicollinearity) was 0.785
    - This shows that the variables we dropped were not affecting the model
- RMSE and MAE values are comparable for train and test sets, indicating that the model is not overfitting.

	Actual Values	Fitted Values	Residuals
731	0.40	0.445434	-0.045434
716	0.70	0.677403	0.022597
640	0.42	0.433999	-0.013999
804	0.55	0.562030	-0.012030
737	0.59	0.547786	0.042214

Figure 36: Checking Residuals

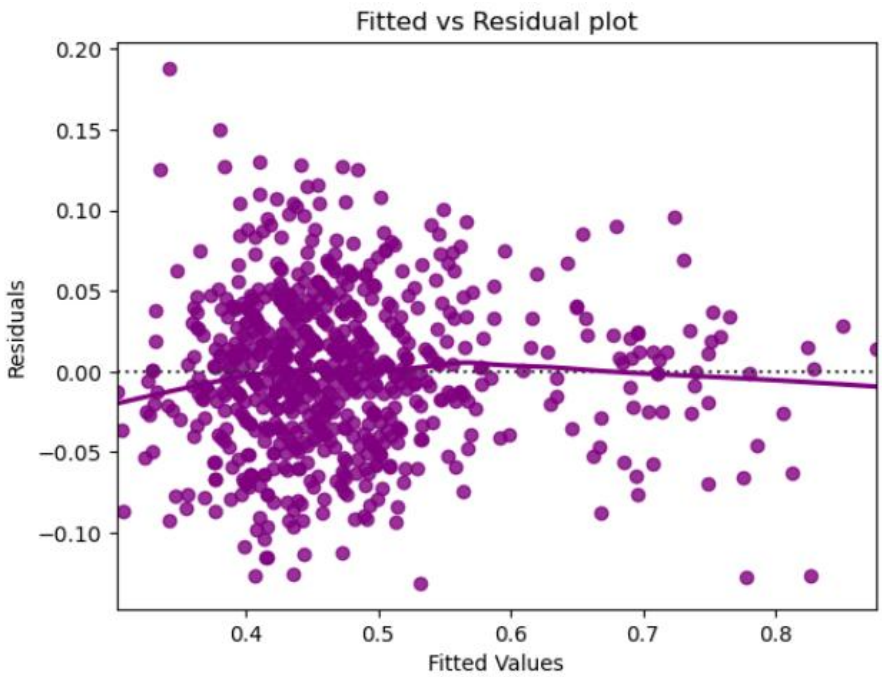

Figure 37: Scatter Plot of Residuals

- The scatter plot shows the distribution of residuals (errors) vs fitted values (predicted values).

- **We see no pattern in the plot above. Hence, the assumptions of linearity and independence are satisfied.**

10.1.3 TEST FOR NORMALITY

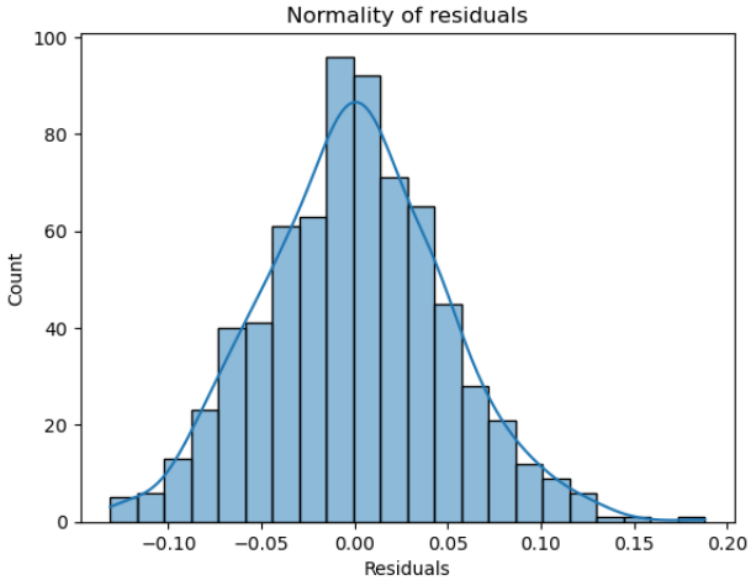

Figure 38: Normality of Residuals

Observations

- The histogram of residuals does have a bell shape.

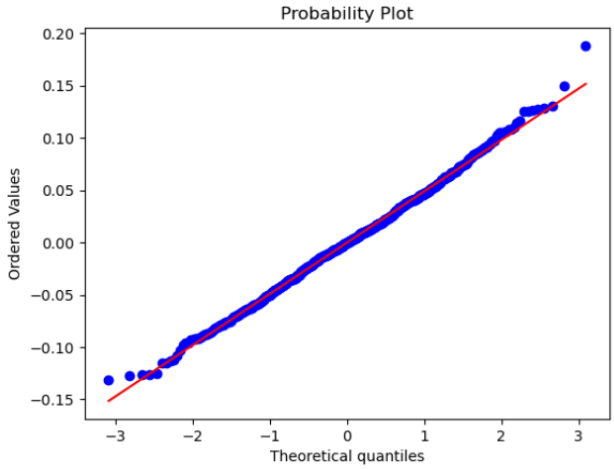

Figure 39: Q-Q Plot of Residuals

Observations

- The residuals more or less follow a straight line except for the tails.

```
ShapiroResult(statistic=0.9973155427169238, pvalue=0.31085896470057883)
```

Figure 40: Shapiro test evidence

- Since p-value > 0.05, the residuals are not normal as per the Shapiro-Wilk test.
- Strictly speaking, the residuals are not normal.
- However, as an approximation, we can accept this distribution as close to being normal.
- **So, the assumption is satisfied.**

10.1.4 TEST FOR HOMOSCEDASTICITY

```
[('F statistic', 1.131361290420075), ('p-value', 0.12853551819087372)]
```

Figure 41: Evidence of Homoscedasticity

Since p-value > 0.05, we can say that the residuals are homoscedastic. So, this assumption is satisfied.

11.0 Predictions on test data

	Actual	Predicted
983	0.43	0.434802
194	0.51	0.500314
314	0.48	0.430257
429	0.41	0.492544
267	0.41	0.487034
746	0.68	0.680000
186	0.62	0.595078
964	0.48	0.503909
676	0.42	0.490313
320	0.58	0.560155

Figure 42: Predictions of Test data

- We can observe here that our model has returned pretty good prediction results, and the actual and predicted values are comparable.

12.0 Final Model summary

```
                            OLS Regression Results
==============================================================================
Dep. Variable:          views_content   R-squared:                       0.789
Model:                            OLS   Adj. R-squared:                  0.786
Method:                 Least Squares   F-statistic:                     233.8
Date:                Sun, 15 Dec 2024   Prob (F-statistic):          7.03e-224
Time:                        19:54:08   Log-Likelihood:                 1120.2
No. Observations:                 700   AIC:                            -2216.
Df Residuals:                     688   BIC:                            -2162.
Df Model:                          11
Covariance Type:            nonrobust
==========================================================================================
                           coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                    0.0747      0.015      5.110      0.000       0.046       0.103
visitors                 0.1291      0.008     16.440      0.000       0.114       0.145
major_sports_event      -0.0606      0.004    -15.611      0.000      -0.068      -0.053
views_trailer            0.0023    5.5e-05     42.414      0.000       0.002       0.002
dayofweek_Monday         0.0321      0.012      2.731      0.006       0.009       0.055
dayofweek_Saturday       0.0570      0.007      8.042      0.000       0.043       0.071
dayofweek_Sunday         0.0344      0.008      4.456      0.000       0.019       0.050
dayofweek_Thursday       0.0154      0.007      2.307      0.021       0.002       0.029
dayofweek_Wednesday      0.0465      0.004     10.532      0.000       0.038       0.055
season_Spring            0.0226      0.005      4.259      0.000       0.012       0.033
season_Summer            0.0434      0.005      8.112      0.000       0.033       0.054
season_Winter            0.0282      0.005      5.362      0.000       0.018       0.039
==============================================================================
Omnibus:                        3.254   Durbin-Watson:                   1.996
Prob(Omnibus):                  0.196   Jarque-Bera (JB):                3.077
Skew:                           0.139   Prob(JB):                        0.215
Kurtosis:                       3.168   Cond. No.                         662.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

Figure 43: Final Model Summary

Training Performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.048841	0.038385	0.788937	0.785251	8.595246

Figure 44: Final Model Training data Performance

Test Performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.051109	0.041299	0.761753	0.751792	9.177097

Figure 45: Test data Performance

- The model is able to explain ~79% of the variation in the data
- The train and test RMSE and MAE are low and comparable. So, our model is not suffering from overfitting

- The MAPE on the test set suggests we can predict within 9.2% of the First day viewership of content.
- Hence, we can conclude the model *olsmodel_final* is good for prediction as well as inference purposes.

13.0 Linear Model Mathematical Equation.

views_content = 0.07467052053721349 + 0.129095818258941 * (visitors) + -0.060555078181373276 * (major_sports_event) + 0.0023308167861640127 * (views_trailer) + 0.032065806790236344 * (dayofweek_Monday) + 0.0570285966016511 * (dayofweek_Saturday) + 0.03438622992362495 * (dayofweek_Sunday) + 0.015449441769973173 * (dayofweek_Thursday) + 0.046494803669848164 * (dayofweek_Wednesday) + 0.02260491581811801 * (season_Spring) + 0.04339100263609974 * (season_Summer) + 0.028230557183976785 * (season_Winter)

14.0 Conclusions and Recommendations

Significant Predictors:

1. **Visitors:** Positive coefficient (0.1291) and a p-value of 0.000, indicating a significant positive impact on views_content.
2. **Major Sports Event:** Negative coefficient (-0.0606) and a p-value of 0.000, indicating a significant negative impact on views_content.
3. **Views Trailer:** Positive coefficient (0.0023) and a p-value of 0.000, indicating a significant positive impact on views_content.
4. **Day of the Week:**
    - Monday, Saturday, Sunday, Thursday, and Wednesday have significant positive coefficients, indicating higher views on these days compared to the reference day.
5. **Season:**
    - Spring, Summer, and Winter have significant positive coefficients, indicating higher views in these seasons compared to the reference season.

Recommendations:

6. **Increase Visitor Engagement:**
    - Since the number of visitors has a significant positive impact on views_content, efforts should be made to increase visitor engagement through marketing and promotional activities.
7. **Leverage Trailers:**
    - The positive impact of views_trailer suggests that promoting trailers can effectively increase content views. Consider investing in high-quality trailers and promoting them across various platforms.
8. **Optimize Content Release Schedule:**
    - Given the significant positive impact of certain days of the week (Monday, Saturday, Sunday, Thursday, and Wednesday), consider scheduling major content releases on these days to maximize views.
9. **Consider Seasonal Trends:**
    - The positive impact of Spring, Summer, and Winter seasons indicates that content views are higher during these periods. Plan major content releases and marketing campaigns around these seasons to capitalize on higher viewership.

10. **Monitor Major Sports Events:**
    - The negative impact of major sports events suggests that viewership decreases during these times. Avoid scheduling major content releases during major sports events to prevent a drop in views.

15.0 Answers to EDA questions

1. What does the distribution of content views look like?

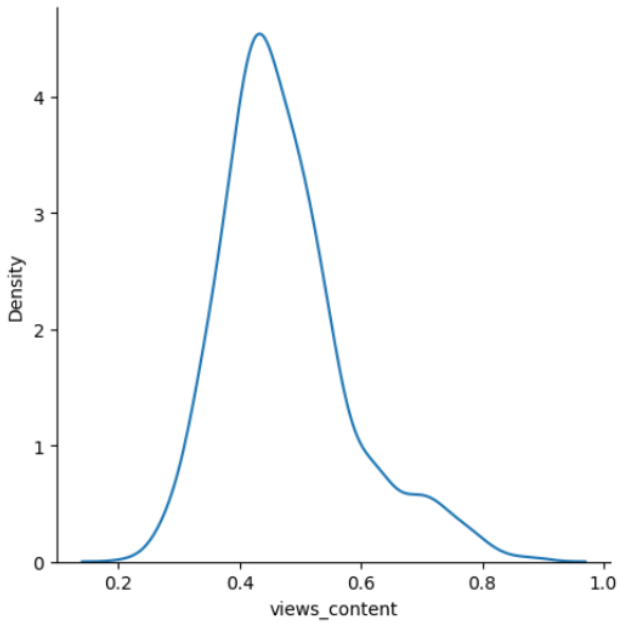

Figure 46: Distribution of views_content attribute

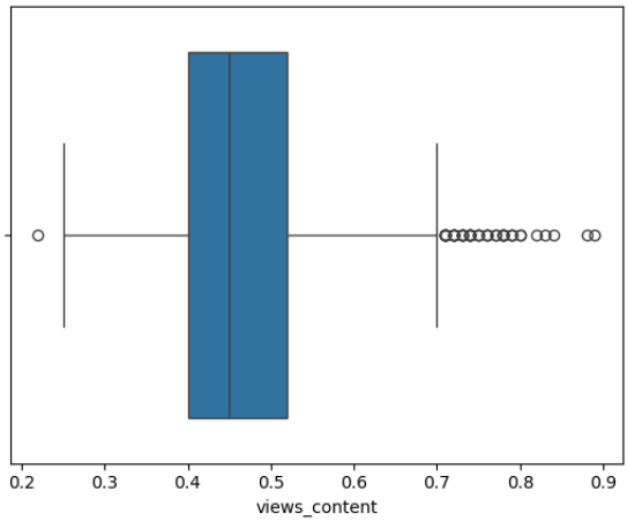

Figure 47: Outliers in views_content attribute

From the above plot below mentioned are the observations

- Data is slightly skewed towards right.
- There are some outliers in this attribute.

2. What does the distribution of genres look like?

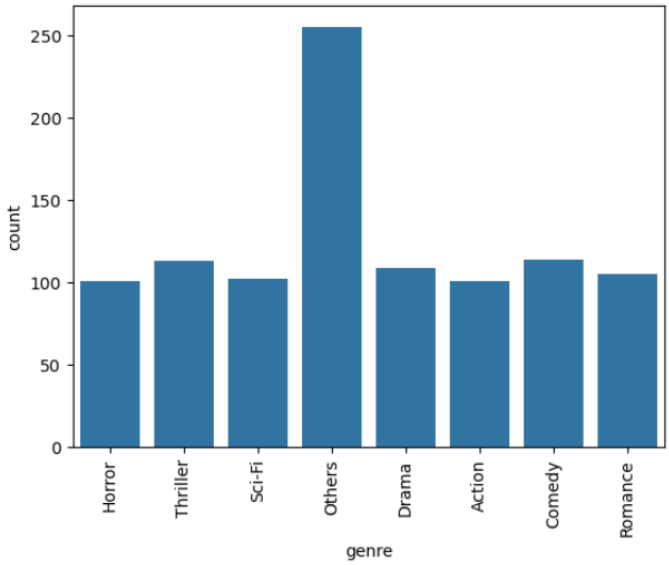

Figure 48: Distribution of genre attribute

- All the genres are having more or less equal count of content. But majority is categorized as others
- Thriller and Comedy Genres have most content next to other genres.

3. The day of the week on which content is released generally plays a key role in the viewership. How does the viewership vary with the day of release?

dayofweek	views_content
Friday	164.83
Monday	11.23
Saturday	43.82
Sunday	32.44
Thursday	45.65
Tuesday	11.22
Wednesday	164.21

Figure 49: Summary of views_content by day of week

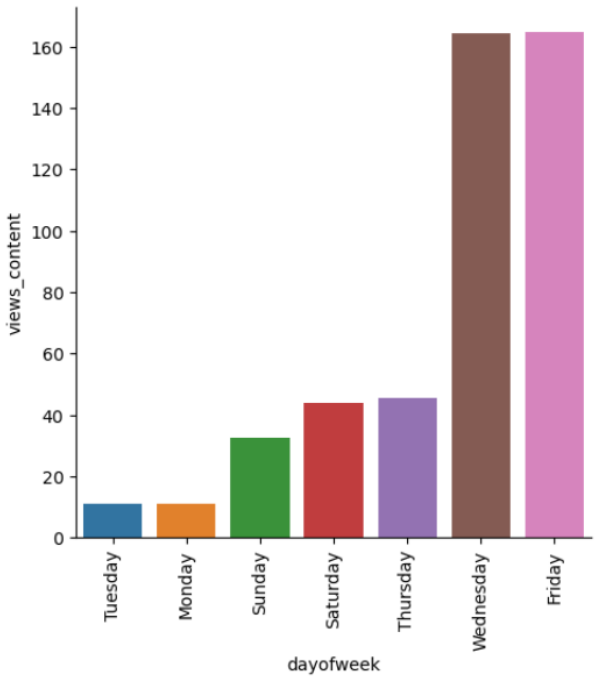

Figure 50: dayofweek vs views_content

Observations on above 2 graphs

- First day views_count is more for the content which release on Wednesday and Friday.
- Content released on Friday has more total number of First day views.
- Content released on Wednesday is in next in line beside Friday wrt to First day total views.
- Content released on Tuesday has least total number of First day views compared to all other days.

4. How does the viewership vary with the season of release?

season	views_content
Fall	112.23
Spring	115.39
Summer	121.22
Winter	124.56

Figure 51: Summary of views_content with season

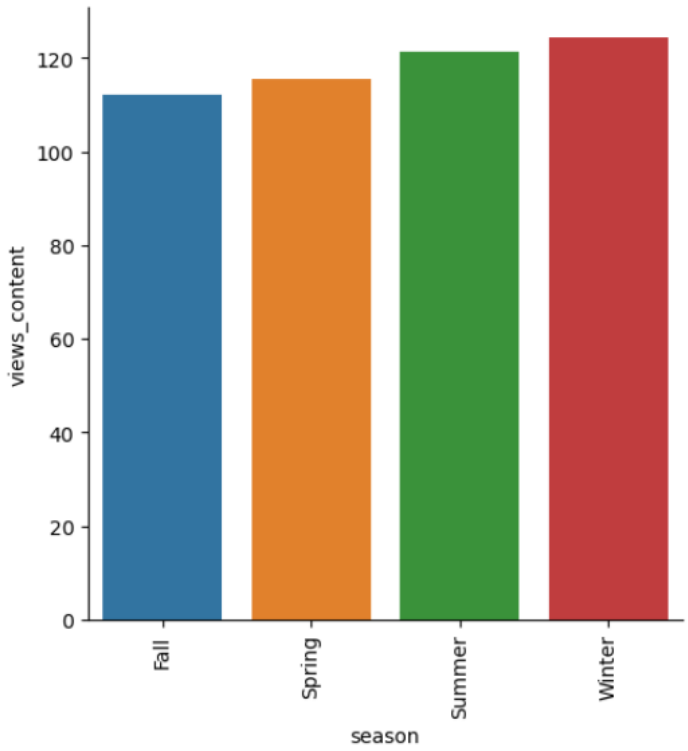

Figure 52: views_content vs season

Observations from above 2 graphs

- Content released in Winter season has higher First day viewership
- Content released in summer season is higher First day viewership next to winter.
- Content released in Fall season has least total First day viewership compared to all the other seasons.

5. What is the correlation between trailer views and content views?

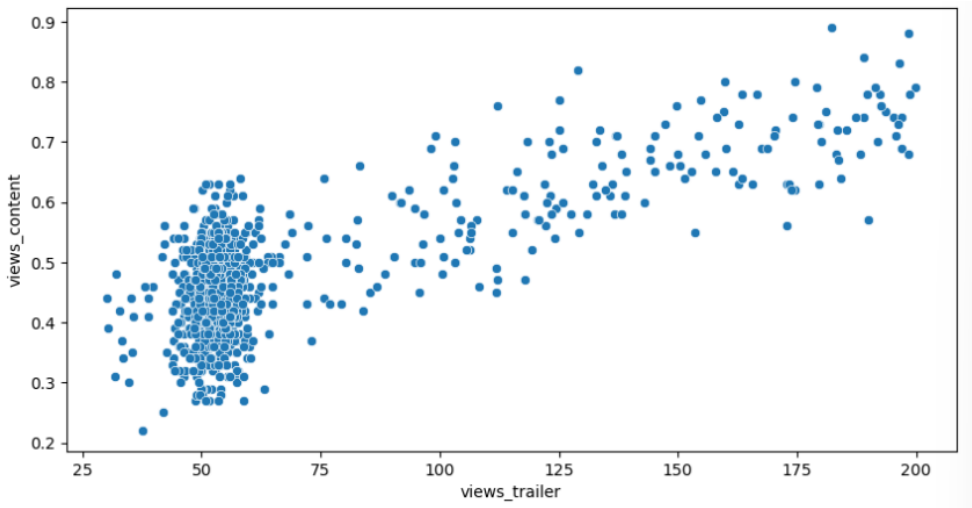

Figure 53: views_trailer vs views_content

- A positive correlation or an increasing trend can be clearly observed between views_trailer and views_content.

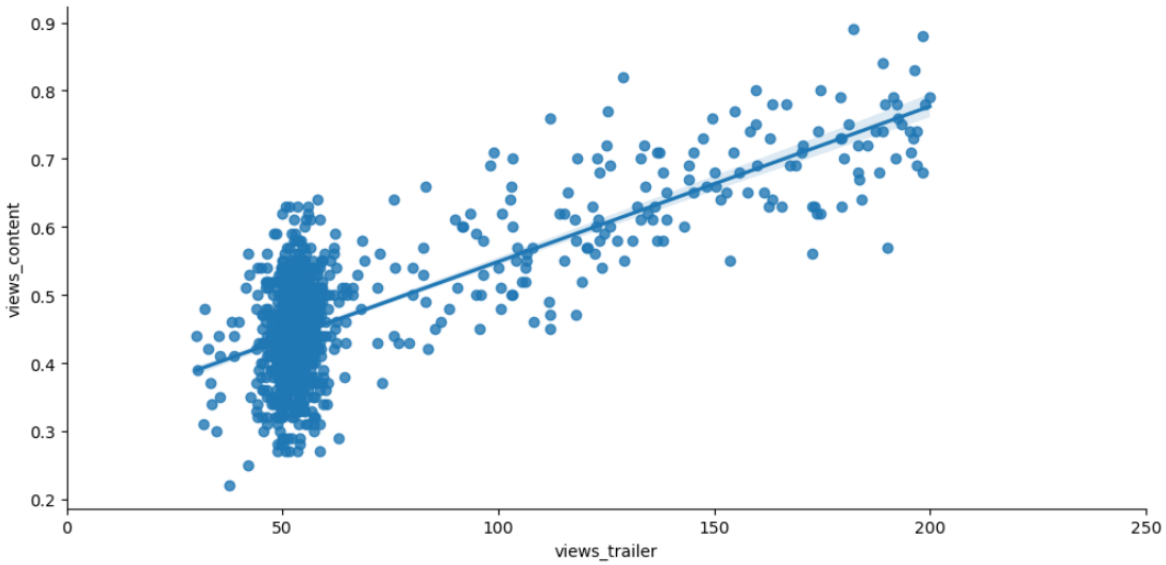

Figure 54: views_trailer vs views_content correlation

with a positive correlation of 0.75 (refer heat map)